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SIT225: Data Capture Technologies

Activity 6.1: Plotly data dashboard

Plotly Dash apps give a point-&-click interface to models written in Python, vastly expanding the notion of what's possible in a traditional "dashboard". With Dash apps, data scientists and engineers put complex Python analytics in the hands of business decision-makers and operators. In this activity, you will learn basic building blocks of Plotly to create Dash apps.

Hardware Required

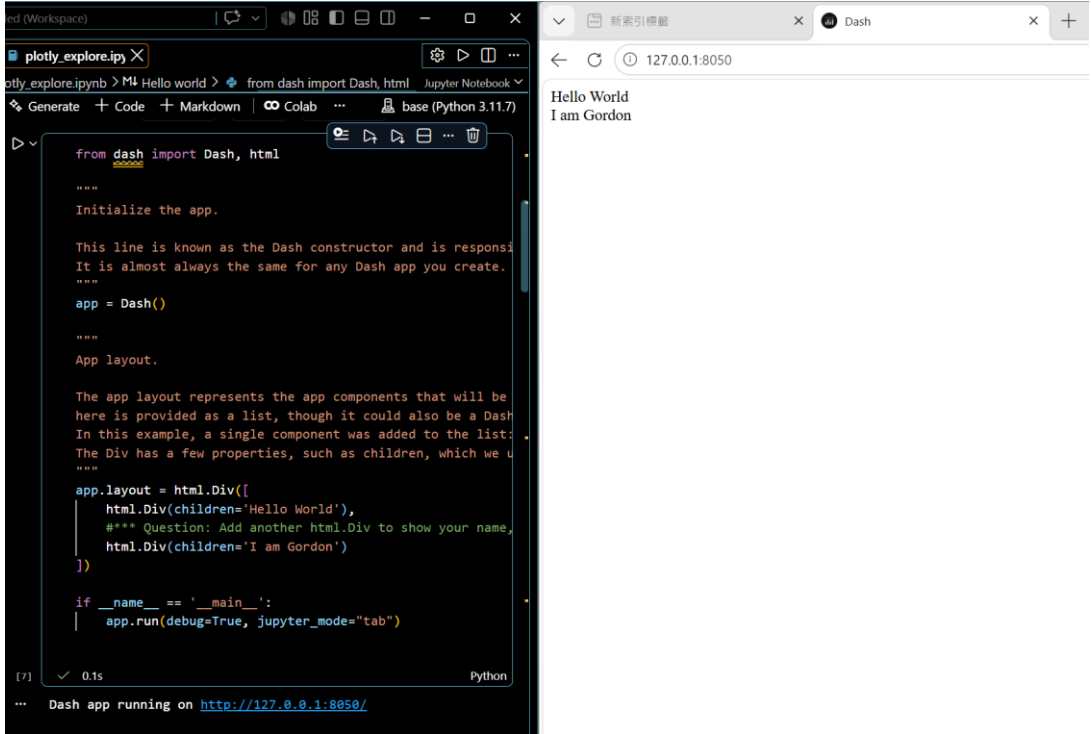
No hardware is required.

Software Required

Plotly library and Dash module
Python 3

Steps

Step	Action
1	Install Plotly and dash using the command below in the command line. <pre>\$ pip install plotly dash</pre> You can download Jupyter Notebook from here (https://github.com/deakin-deep-dreamer/sit225/blob/main/week_6/plotly_explore.ipynb) and run all the cells. The Notebook contains multiple sections such as Hello World which follows a sample code in a following cell. If you run the Hello world cell it will show Plotly Dash web page. The cell also includes a Question (**** Question) which you will need to carry out to get a modified output. You will need to capture the output and share the screenshot in the following steps.

2	Question: Hello World cell has a question - add another html.Div to show your name, and re-run the cell for output. You will need to update the code, run the cell, capture the screenshot of the output and paste it here. Answer:  The screenshot shows a Jupyter Notebook on the left and a web browser on the right. The notebook cell contains Python code for a Dash application. The code imports Dash and html, initializes the app, and sets the layout to a Div containing two other Divs: one with 'Hello World' and another with 'I am Gordon'. The code also includes a comment: '*** Question: Add another html.Div to show your name,'. The browser on the right shows the output of the Dash application at the URL 127.0.0.1:8050, displaying 'Hello World' and 'I am Gordon'.
3	Question: Connecting to Data cell has a question - change page size and observe the change in widget controls such as, total number of pages. You will need to update the code, run the cell, capture the screenshot of the output and paste it here. Answer:

```
Explorer (Ctrl+Shift+I) nb
C:\Users\gordo> OneDrive> Desktop> Y3> T2> SIT225 Data Capture> SIT225 2026T2> Web Jupyter Notebook
Generate + Code + Markdown | Colab | Run All | Restart ... base (Python 3.11.7)

Connecting to Data

There are many ways to add data to an app: APIs, external databases, local .txt files, JSON
files, and more. In this example, we will highlight one of the most common ways of
incorporating data from a CSV sheet.

# Import packages
# We import the dash_table module to display the data inside a Dash DataTable.
from dash import Dash, html, dash_table

# We also import the pandas library to read the CSV sheet data.
import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=10)
])

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode='tab')
```

My First App with Data

country	pop	continent	lifeExp	gdpPerCap
Afghanistan	31889923	Asia	43.828	974.5803384
Albania	3600523	Europe	76.423	5937.629525999999
Algeria	33333216	Africa	72.301	6223.367465
Angola	12428476	Africa	42.731	4797.231267
Argentina	40301927	Americas	75.32	12779.37964
Australia	20434176	Oceania	81.235	34435.367439999995
Austria	8199783	Europe	79.829	36126.4927
Bahrain	708573	Asia	75.635	29796.04834
Bangladesh	150448339	Asia	64.062	1391.253792
Belgium	10392226	Europe	79.441	33692.60508

```
plotly_explore.ipynb
SIT225 2026T2> Web 6> plotly_explore.ipynb> M4 Connecting to Data> # Import packages Jupyter Notebook
Generate + Code + Markdown | Colab | Run All | Restart ... base (Python 3.11.7)

# Import packages
# We import the dash_table module to display the data inside a Dash DataTable.
from dash import Dash, html, dash_table

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import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=5)
])

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode='tab')
```

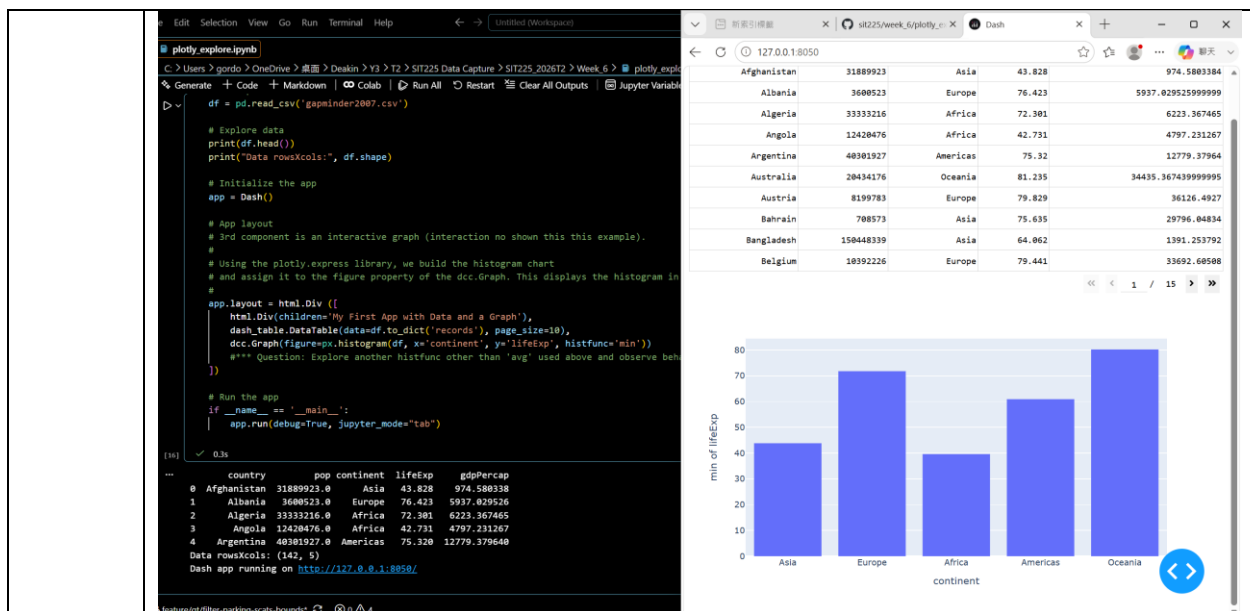
My First App with Data

country	pop	continent	lifeExp	gdpPerCap
Afghanistan	31889923	Asia	43.828	974.5803384
Albania	3600523	Europe	76.423	5937.629525999999
Algeria	33333216	Africa	72.301	6223.367465
Angola	12428476	Africa	42.731	4797.231267
Argentina	40301927	Americas	75.32	12779.37964

4

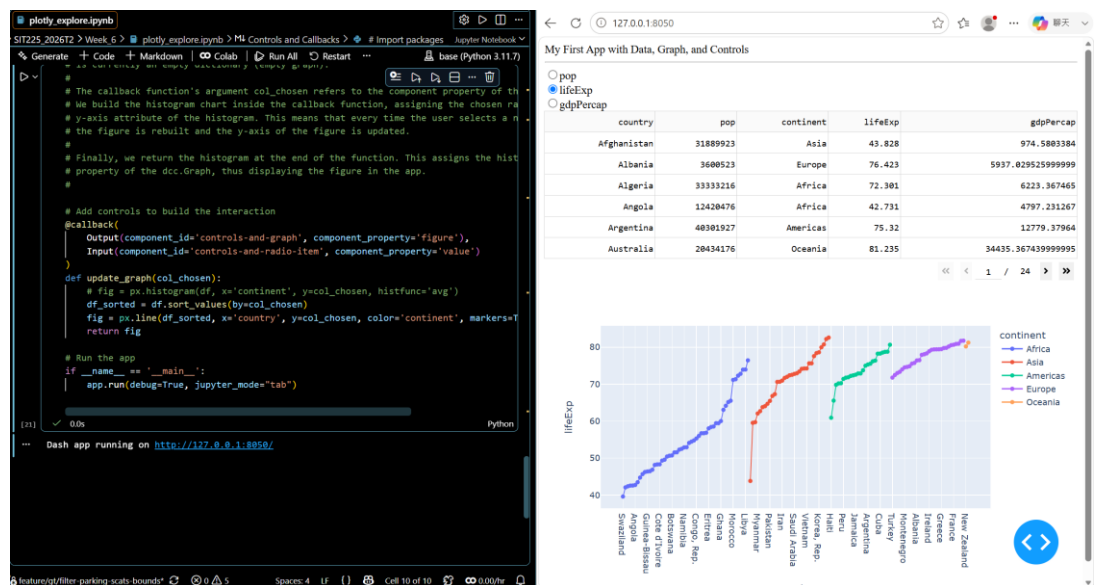
Question: Visualising data cell has a question - explore another histfunc other than 'avg' used above and observe behaviour. You will need to update the code, run the cell, capture the screenshot of the output and paste it here.

Answer:



5 **Question:** Controls and Callbacks cell has a question - use line graphs instead of histogram. You will need to update the code, run the cell, capture the screenshot of the output and paste it here.

Answer:



6 **Question:** Now you have learned how to use Plotly Dash for visualising your data, describe how you will be using this tool for your desired sensor

	monitoring dashboard with a number of sensors including DHT22 or accelerometer data. Answer: plotly dash is a tool to visualize data, since it manage by code base, I can use capture sensors data by Arduino cloud or python script then pass that data to dash so I can see the data in real time. And find out more patten.
7	Question: Convert the Notebook to PDF and merge with this activity sheet PDF. You will need this merged PDF to combine with this week's OnTrack task for submission.

```
# Fill in student ID and name
#
student_id = "s223532749"
student_first_last_name = "Gordon Tse"
print(student_id, student_first_last_name)

s223532749 Gordon Tse

# install plotly and dash, if not yet already
# ! pip install plotly dash

import plotly, dash
print(plotly.__version__)
print(dash.__version__)

5.9.0
2.14.2
```

Hello world

Building and launching an app with Dash can be done with just 5 lines of code. Follow the tutorial (<https://dash.plotly.com/tutorial>) for more detail.

```
from dash import Dash, html

"""
Initialize the app.

This line is known as the Dash constructor and is responsible for
initializing your app.
It is almost always the same for any Dash app you create.
"""
app = Dash()

"""
App layout.

The app layout represents the app components that will be displayed in
the web browser and
here is provided as a list, though it could also be a Dash component.
In this example, a single component was added to the list: an
html.Div.
The Div has a few properties, such as children, which we use to add
text content to the page: "Hello World".
"""
app.layout = html.Div([
    html.Div(children='Hello World'),
    #*** Question: Add another html.Div to show your name, and re-run
```

```

the cell for output.
    html.Div(children='I am Gordon')
])

if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")

Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

```

Connecting to Data

There are many ways to add data to an app: APIs, external databases, local .txt files, JSON files, and more. In this example, we will highlight one of the most common ways of incorporating data from a CSV sheet.

```

# Import packages

# We import the dash_table module to display the data inside a Dash
# DataTable.
from dash import Dash, html, dash_table

# We also import the pandas library to read the CSV sheet data.
import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=5)
    #*** Question: Change page size and observe the change in widget
    # controls
    # such as, total number of pages.
])

```

```
# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")
```

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923.0	Asia	43.828	974.580338
1	Albania	3600523.0	Europe	76.423	5937.029526
2	Algeria	33333216.0	Africa	72.301	6223.367465
3	Angola	12420476.0	Africa	42.731	4797.231267
4	Argentina	40301927.0	Americas	75.320	12779.379640

Data rowsXcols: (142, 5)
Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

Visualising data

The Plotly graphing library has more than 50 chart types to choose from. In this example, we will make use of the histogram chart.

```
# Import packages

# We import the dcc module (DCC stands for Dash Core Components).
# This module includes a Graph component called dcc.Graph, which is
# used to render interactive graphs.
from dash import Dash, html, dash_table, dcc

# We also import the plotly.express library to build the interactive
# graphs.
import plotly.express as px

import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout
# 3rd component is an interactive graph (interaction not shown in this
# example).
#
# Using the plotly.express library, we build the histogram chart
```

```

# and assign it to the figure property of the dcc.Graph. This displays
the histogram in our app.
#
app.layout = html.Div ([
    html.Div(children='My First App with Data and a Graph'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=10),
    dcc.Graph(figure=px.histogram(df, x='continent', y='lifeExp',
histfunc='min'))
    *** Question: Explore another histfunc other than 'avg' used
above and observe behaviour.
])

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")

```

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923.0	Asia	43.828	974.580338
1	Albania	3600523.0	Europe	76.423	5937.029526
2	Algeria	33333216.0	Africa	72.301	6223.367465
3	Angola	12420476.0	Africa	42.731	4797.231267
4	Argentina	40301927.0	Americas	75.320	12779.379640

Data rowsXcols: (142, 5)
Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

Controls and Callbacks

So far you have built a static app that displays tabular data and a graph. However, as you develop more sophisticated Dash apps, you will likely want to give the app user more freedom to interact with the app and explore the data in greater depth. To achieve that, you will need to add controls to the app by using the callback function.

In this example we will add radio buttons to the app layout. Then, we will build the callback to create the interaction between the radio buttons and the histogram chart.

```

# Import packages

# We import dcc like we did in the previous section to use dcc.Graph.
# In this example, we need dcc for dcc.Graph as well as the radio
buttons component, dcc.RadioItems.
#
# To work with the callback in a Dash app, we import the callback
module and the two arguments
# commonly used within the callback: Output and Input.
#
from dash import Dash, html, dash_table, dcc, callback, Output, Input

```

```

import pandas as pd
import plotly.express as px

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Initialize the app
app = Dash()

# App layout
app.layout = html.Div([
    html.Div(children='My First App with Data, Graph, and Controls'),
    html.Hr(),
    dcc.RadioItems(options=['pop', 'lifeExp', 'gdpPercap'],
value='lifeExp', id='controls-and-radio-item'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=6),
    dcc.Graph(figure={}, id='controls-and-graph')
    *** Question: Use line graphs instead of histogram.
])

#
# Notice that we add that RadioItems component to the layout, directly
# above the DataTable.
# There are three options, one for every radio button.
# The lifeExp option is assigned to the value property, making it the
# currently selected value.
#
# Both the RadioItems and the Graph components were given id names:
# these will be used by
# the callback to identify the components.
#
# The inputs and outputs of our app are the properties of a particular
# component.
# In this example, our input is the value property of the component
# that has the ID "controls-and-radio-item".
# If you look back at the layout, you will see that this is currently
# lifeExp.
# Our output is the figure property of the component with the ID
# "controls-and-graph", which
# is currently an empty dictionary (empty graph).
#
# The callback function's argument col_chosen refers to the component
# property of the input lifeExp.
# We build the histogram chart inside the callback function, assigning
# the chosen radio item to the
# y-axis attribute of the histogram. This means that every time the
# user selects a new radio item,
# the figure is rebuilt and the y-axis of the figure is updated.
#
# Finally, we return the histogram at the end of the function. This

```

```

assigns the histogram to the figure
# property of the dcc.Graph, thus displaying the figure in the app.
#

# Add controls to build the interaction
@callback(
    Output(component_id='controls-and-graph',
            component_property='figure'),
    Input(component_id='controls-and-radio-item',
           component_property='value')
)
def update_graph(col_chosen):
    # fig = px.histogram(df, x='continent', y=col_chosen,
histfunc='avg')
    df_sorted = df.sort_values(by=col_chosen)
    fig = px.line(df_sorted, x='country', y=col_chosen,
                  color='continent', markers=True)
    return fig

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")

Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

```